

SAI TARRUN PITTA

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PROFESSIONAL SUMMARY

Software Engineer with 3+ years of experience designing and delivering scalable backend systems, distributed cloud platforms, and AI-enabled applications. Skilled in building high-volume payment platforms, cloud infrastructure, RAG retrieval pipelines, and secure production systems, with a strong focus on reliability, performance optimization, and security-first engineering.

TECHNICAL SKILLS

Languages:

Python, Go, JavaScript, TypeScript, SQL, Java, C++

Frontend:

React, Next.js, Tailwind CSS, Redux

Backend:

Node.js, FastAPI, Django, Express, REST APIs, gRPC

Databases:

PostgreSQL, MongoDB, DynamoDB, Redis, Elasticsearch, Pinecone

Cloud & DevOps:

AWS, GCP, Docker, Kubernetes, Terraform, GitHub Actions

AI/ML:

TensorFlow, scikit-learn, LangChain, RAG pipelines, LLMs

PROFESSIONAL EXPERIENCE

Pacific Life — Software Engineer Intern

May 2025 - Dec 2025

Newport Beach, CA

- Automated 98% of claims triage workflows with serverless ML/LLM service (AWS Lambda, TensorFlow, LangChain)
- Reduced claims processing time by 75% and failures by 30% with human-in-the-loop validation checkpoints
- Accelerated policy lookup by 98% using RAG pipeline with Pinecone semantic search and access-controlled retrieval
- Built TypeScript API layer with Redis caching, reducing dashboard load time 50% (4s → 2s)

Accenture — Software Engineer

Jul 2022 - Jun 2024

Hyderabad, India

- Scaled gRPC microservice to process 2M+ daily transactions with 99.99% uptime using Protocol Buffers and Redis locking
- Optimized AWS infrastructure (EKS, ElastiCache) reducing compute 50% and cloud spend 35% (\$15K/month)
- Optimized SQL queries reducing latency 60% and database CPU usage 73%
- Designed reliability architecture with CloudWatch/Prometheus/Splunk, reducing MTTR 25x

Cognizant — Software Engineer

May 2021 - Apr 2022

Hyderabad, India

- Engineered ML churn prediction system with TensorFlow/React dashboards, achieving 12% churn reduction
- Developed scalable REST APIs for payment analytics with NLP-based risk scoring
- Secured payment application with OpenID Connect/JWT, reducing unauthorized access risk 35%
- Automated CI/CD workflows using Docker/GitHub Actions, reducing release lead time 50%

EDUCATION

California State University, Fullerton

Aug 2024 - May 2026

Master of Science, Computer Science — Fullerton, CA

- Publication: "Hardware Trojan Detection with ML and Power Side Channels" (IEEE CNS 2025)
- Coursework: Advanced Algorithms, Cloud Computing, System Design, Machine Learning, AI, Distributed Systems

PROJECTS

LLM-Powered Knowledge Retrieval Platform

Distributed knowledge retrieval with semantic search, vector indexing, and NLP query matching (25% accuracy improvement, 40% faster search)

- LangChain, Pinecone, FastAPI, React, TypeScript, Redis

Production Portfolio & Personal Website

Performance-optimized portfolio with security-first engineering and comprehensive code quality tooling

- React, Vite, Tailwind CSS, TypeScript strict mode, ESLint, Prettier, Git hooks

Event-Driven E-Commerce Modernization

Monolith-to-microservices migration using Strangler pattern with zero-downtime canary deployments

- Node.js, Kafka, PostgreSQL, Docker, Kubernetes, AWS